The vibes that I summoned…

Thoughts on working with AI agents.

Chris Pahl · 2026

The Spectrum (on Reddit)

Full manual
every keystroke yours

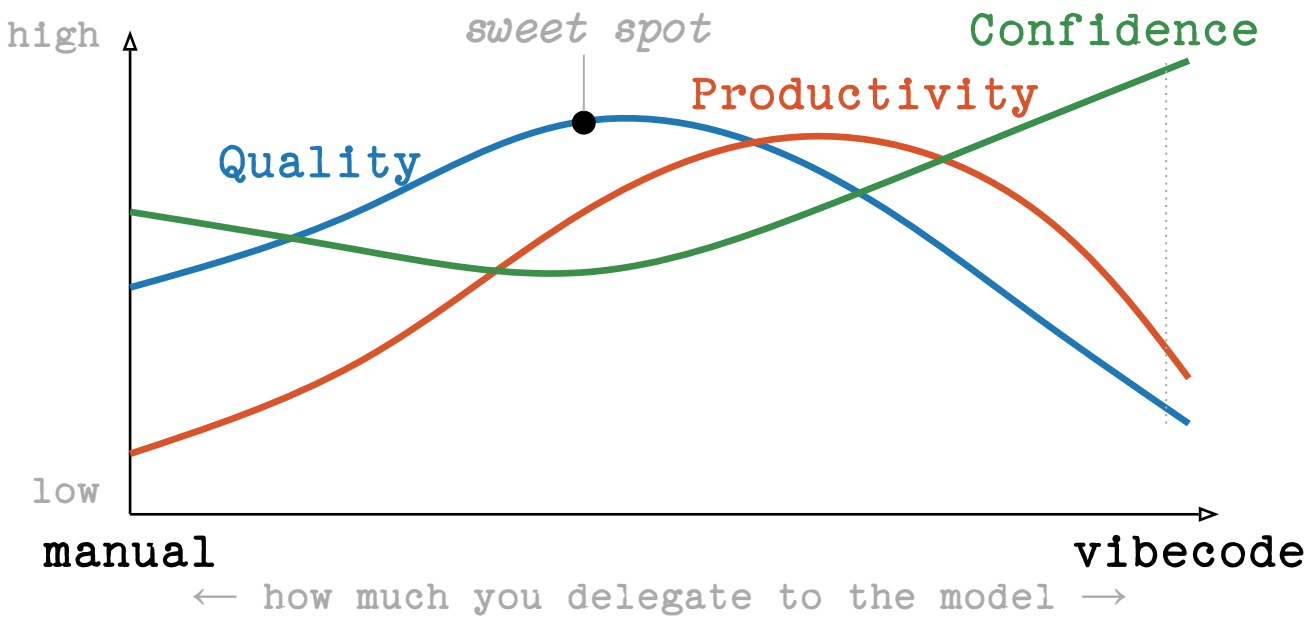
Full vibecode
just make it work

⟵ more control · more understanding | more productivity · more delegation ⟶

Using GenAI is a tradeoff between control and productivity.

It’s not black & white.

My take: Quality vs Productivity



Quality peaks in the middle. Productivity follows it down. Confidence diverges from both -> Dunning-Kruger zone!

13%

But there's already data:

-19%

slower with AI

experienced devs · own mature repos · predicted +24%

METR · RCT · 2025

~40%

insecure programs

Copilot output across 89 security-relevant scenarios

NYU CCS · 2021

8×

more duplicated code

block-level clones up · refactoring 24% → 10%

GitClear · 211M LoC · 2024

-7.2%

delivery stability

drop with AI adoption · 39% distrust AI code

DORA / Google · 2024

29.5%

Python with CWEs

AI-generated snippets containing known vulnerabilities

Fu et al. · ACM TOSEM · 2025

↓

trust = less scrutiny

more trust in GenAI ⟹ less critical thinking applied

Lee et al. · Microsoft + CMU · CHI 2025

Good use cases

rubber-ducking ideation

explain unfamiliar code test generation

pair programming boilerplate refactoring assist

learning accelerator documentation drafts naming things

summarisation regex / SQL crafting edge-case brainstorming

translation code review companion mock data / fixtures

error decoding onboarding ramp-up log analysis proofreading

automation

Risks for all

ecological cost

Convincing hallucinations

schools

licensing

Atrophy

concentration

misinformation at scale

Prompt injection & MCP

content degradation

No juniors hired

hardware crisis

operator bias

recursive collapse

economic transfer

data privacy

cyberattack automation

communities thinning out

Risks for us

ecological cost

Convincing hallucinations

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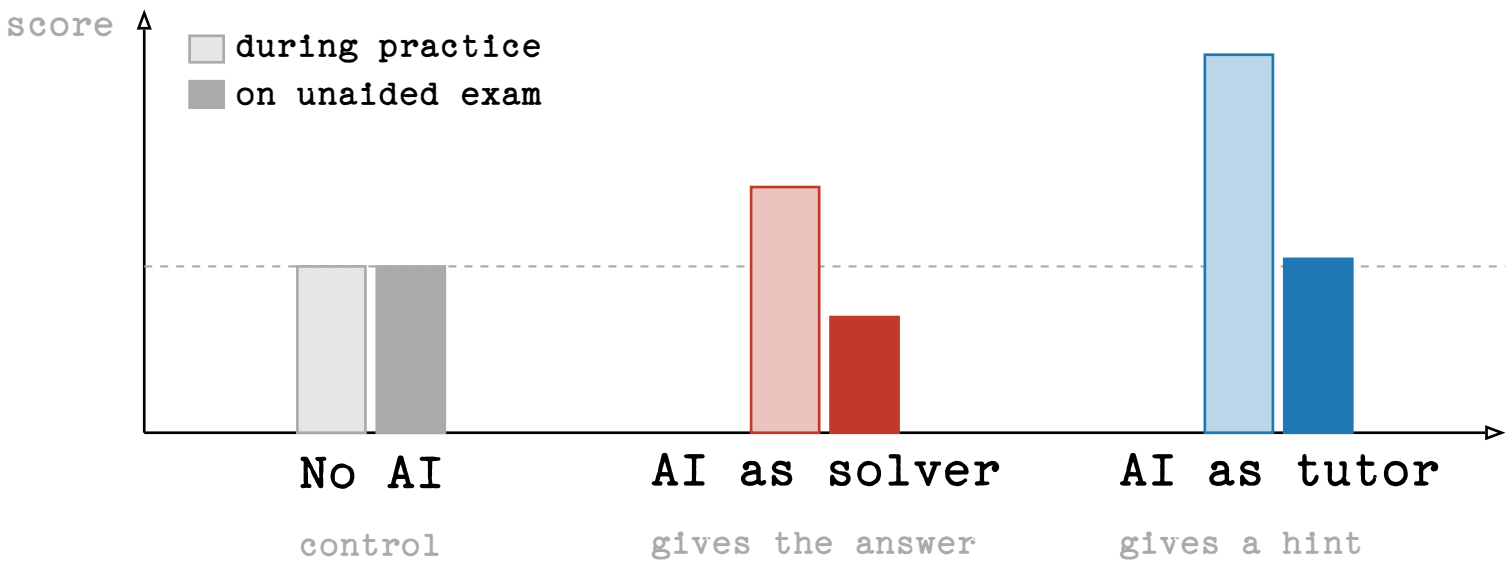
Exhibit A: schools.

Maybe it's not so much about the tool —
but about how we use it?

33%

Same tool, opposite outcomes



Bastani et al., *Generative AI Can Harm Learning*, SSRN 2024

Are *you* team tutor or team solver?

Keeping up with Claude

Our team's Claude Code account · April 2026

98.7%

suggestions accepted as-is

1.3% rejected. Is that a review process?

27,757

lines accepted last month

≈ 925 LoC/day · careful review ≈ 100–200 LoC/hour

Reviewing 27,757 lines carefully ≈ 138 h. A working month ≈ 176 h.

Either we're 7× faster than the industry — or we didn't review.

Human cognitive biases

Automation bias

We accept machine suggestions more readily than human ones.

click accept · review later · maybe

Anchoring

The first suggestion shapes the solution space — even when it's wrong.

the model frames the problem before you do

Confirmation bias

We hear what we already believe.

the prompt contains the answer we want

Authority bias

Eloquent + fast + confident reads as expert.

fluency ≠ correctness

Dunning–Kruger

We mistake competence we observe for competence we have.

"this is what I would have written"

Illusion of explanatory depth

We think we understand — until asked to explain.

"yes, I read the diff" → bug

Statistical model biases

Sycophancy

It's easy to talk the model out of a correct answer.

echo chambers — but for code

Historical / representation bias

Trained on the web — heavily modern, English, Western.

defaults track what's common, not what's right

Attention bias

First and last things dominate; the middle evaporates.

rules buried in long context get ignored

Omission bias

Rare and novel answers get suppressed by common ones.

the model is bad at being weird

Hen & Egg

- If you don’t know what good software looks like -
  how do you write the right prompt?
- If you can’t understand what the model just generated -
  how do you verify it?
- If you can’t code (anymore) -
  how can you understand the diffs?

Humans are still very much required.

How do *you* keep up with Claude?

Five habits that helped me

1. Think first, generate then.
2. Split the work - keep some of it manual.
3. Have a real verification strategy.
4. You only get replaced if you make yourself replaceable.
5. Treat the AI as a colleague.

1. Think first, generate then

Don’t let the model set the anchor.

- Sketch the SQL query yourself,
- Then ask the model to review and optimise.
- Write the function signature and the docstring.
- Then let the model fill the body.
- You set the anchor; the model refines.

2. Split the work

The parts you write are the parts you understand.

- Leave the manual labor work to the model (e.g. build systems)
- Keep doing at least some smaller tasks yourself.
- I suggest: Do most of critical production code yourself.
- Tests, Build system, scripts, boilerplate: Default to Claude.

3. Have a real verification strategy

- Before you accept a generated change, name the signal that would tell you it’s wrong.
- Find other ways if the diff is too big:
  - Property-based tests, fuzzing (e.g. *noise* for Dart).
  - Type checkers, linters, static analysers.
  - Replay against real production data.
- If you can’t say how you’d notice the bug, you don’t have a verification strategy - you have hope.

Example: Noise library for Dart. I don’t speak Dart.

4. You only get replaced if you make yourself replaceable.

You only get replaced if you make yourself replaceable.

- The replaceable parts of your job are the parts where you're already a slow autocomplete.
- Spend the time AI saves you on the parts no model can do: judgement, context, design.
- Code was the source of truth before, that's changing.
- Now you have more time to think about design and purpose.
- If you just operate Claude you are replaceable.

5. Treat the AI as a colleague

Not an oracle, not a junior, not magic.

- Push back. Ask for alternatives. Disagree.
- It’s a very eager, sometimes rather junior colleague with seemingly infinite power.
- If you wouldn’t accept a colleague’s PR with the same reasoning, don’t accept the model’s.

Extra: Software as Theory Building

Peter Naur, 1985

- The code is a by-product. What you're really building is the theory – your team's shared mental model of the problem.
- Documentation captures the artefact, not the theory. The bugs, the dead ends, the "why not this?" decisions live in people's heads.
- This is why handovers fail. The theory doesn't transfer cleanly.

Peter Naur, 1928–2016
Photo: Wikimedia Commons

What that means for AI

- The model has no theory. It has the artefact and the context you give it.
- “Write tests for this file” → it freezes today’s behaviour, bugs included.
- Your job is still to build the theory. The AI helps you write the code that follows from it.

Practical:

- Keep a design doc / decision log. The why, not just the what.
- CLAUDE.md is size-limited — use it for headlines, link out for depth.

Questions or Disagreements?

Homework:

Read SOFTWARE AS THEORY BUILDING.